Exploring NYC Taxi Trips: Time Patterns, Location Hotspots, Airport Travel Times and Dimensionality Reduction

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Abstract

We analyse the 1 458 644 trips of the Kaggle *NYC Taxi Trip Duration* training set (January–June 2016). After removing 1.4 % of the trips as errors (impossible passenger counts, coordinates outside New York, durations below one minute or above three hours, near-zero distances and implausible speeds), we generate distance, speed and calendar features and study (i) time patterns, (ii) spatial patterns with heat-maps and DBSCAN hotspots, (iii) travel times from the Empire State Building to JFK and Newark, and (iv) the structure of a 20 000-trip sample with PCA, t-SNE and UMAP. Demand peaks on Friday and Saturday (statistically tied) and at 18:00–21:00 on weekdays; only Memorial Day changes the daily volume significantly. Hotspots are compact street corners that differ between Friday night (downtown nightlife) and Thursday evening (airports and Midtown hubs). A trip to JFK takes a median 46 min (about 60 min in the afternoon rush, 29 min at night) against 33 min to Newark. Non-linear embeddings reveal seven day-of-week groups and separate airport groups that PCA does not resolve; their clusters are recoverable from the original features (96 % kNN accuracy) but are not separated there, and their number, size and arrangement depend on the seed and hyper-parameters.

Overview and reproducibility

All code, with the question numbers marked, is in the accompanying Jupyter notebook `DVE_Assignment_2.ipynb`; every figure below is produced by it. We use a fixed seed (42) throughout. The Winter Storm of 23–25 January 2016 (only 1 614 trips on Saturday 23 January against about 8 400 on a normal Saturday) is excluded from "typical-day" baselines in Sections 3 and 4. Hand-entered landmark coordinates are used only to *describe* patterns and are approximate.

1 Data cleaning

We computed the great-circle (haversine) distance between pickup and dropoff and the average speed (distance divided by duration); note that haversine is a lower bound on the road distance, so thresholds are generous. Figures 1–2 show the raw data and Table 1 the effect of each rule. The audit found no missing values, no duplicate ids and that `trip_duration` equals dropoff − pickup time for every row.

Preprint. Under review.

- **Passenger count** $\notin [1, 6]$: zero passengers is not a paid trip and 7–9 exceed any taxi's capacity (driver-entered data-entry errors).
- **Location**: the raw coordinates reach longitude -121.9, latitude 51.9 (west coast, Canada, Atlantic). We keep trips whose pickup *and* dropoff lie in lon $[-74.30, -73.65]$, lat $[40.49, 40.92]$, which covers all five boroughs plus JFK and Newark.
- **Duration** < 60 s (cancelled/accidental meter starts; 1 984 trips are shorter than 10 s) and > 3 h (meter left running; up to 40.8 days).
- **Distance** < 100 m: below GPS accuracy in Manhattan's urban canyons; the trip did not go anywhere and the speed is undefined.
- **Speed** > 100 km/h (physically impossible as an *average* in a city, even allowing for haversine's under-estimate) and < 1 km/h (slower than walking, i.e. a stationary vehicle with the meter running).

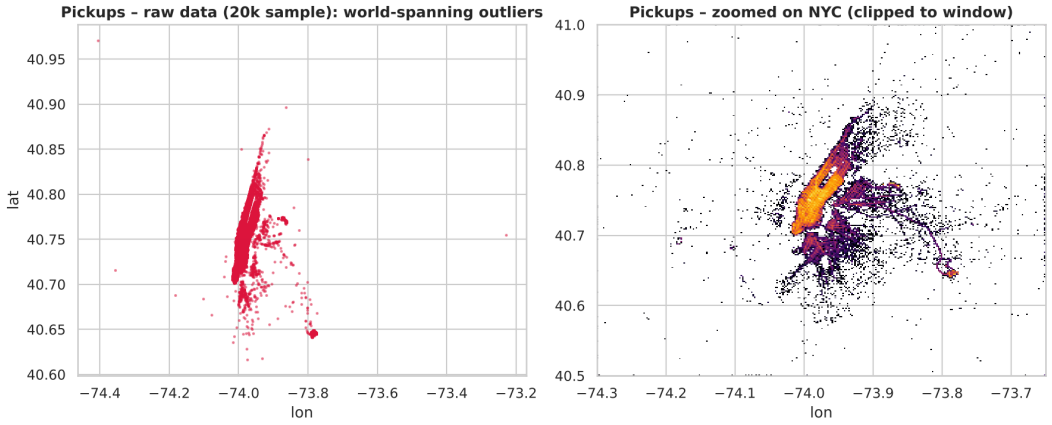


Figure 1: Raw pickups: world-spanning outliers (left) versus the NYC window (right, log density).

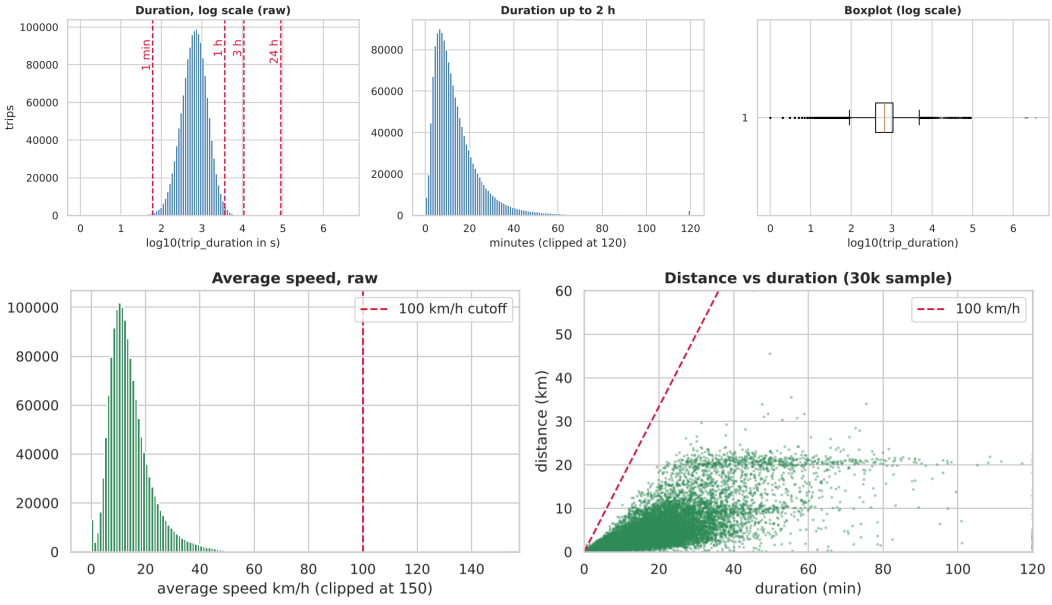


Figure 2: Raw duration (log scale) and average speed; red dashed lines are our cut-offs.

Table 1: Cleaning funnel. "Flagged" = trips breaking the rule on its own; "removed" = additionally removed when rules are applied in this order. In total 20 448 trips (1.40 %) are removed, leaving 1 438 196.

Rule	Flagged	Removed (sequential)
passenger count not in $[1, 6]$	65	65
coordinates outside NYC window	1 096	1 095
duration $<$ 60 s	8 595	8 401
duration $>$ 3 h	2 112	2 104
distance $<$ 100 m	13 319	7 442
speed $>$ 100 km/h	175	33
speed $<$ 1 km/h	13 241	1 308

2 Feature generation

Besides the three required features – **distance of trip** (haversine, km), **day of week** and **average speed** (km/h) – we generated pickup hour, date, month, weekend flag, duration in minutes, the cyclic hour encoding $(\sin \frac{2\pi h}{24}, \cos \frac{2\pi h}{24})$, a Manhattan-grid ($L_1$) distance (median 1.33$\times$ the straight-line distance, as expected on a street grid), the initial compass bearing, and the distances of the pickup and dropoff to the city centre $(40.716662, -74.009899)$. The cleaned data have median distance 2.12 km, median duration 11.1 min and median speed 12.9 km/h.

3 Time-based analysis

The data cover 182 days, so each weekday occurs exactly 26 times and weekday totals are directly comparable.

3.1 Most popular day of the week

Friday and Saturday are the two busiest days and are statistically tied. Friday has the highest total (220 378 pickups), followed by Saturday (217 740), Thursday (215 509), Wednesday (207 384), Tuesday (199 969), Sunday (192 396) and Monday (184 820, the quietest; Friday is 19 % busier). However, the blizzard fell on a Saturday/Sunday; excluding 23–25 January, Saturday leads (8 645 vs 8 476 trips per day) and a Welch t-test on daily counts gives $p = 0.37$, so the difference is not significant. Friday is among the three busiest days in all six months. Thursday–Saturday are the going-out days on top of the commute, whereas Sunday and Monday lack a strong leisure or start-of-week demand.

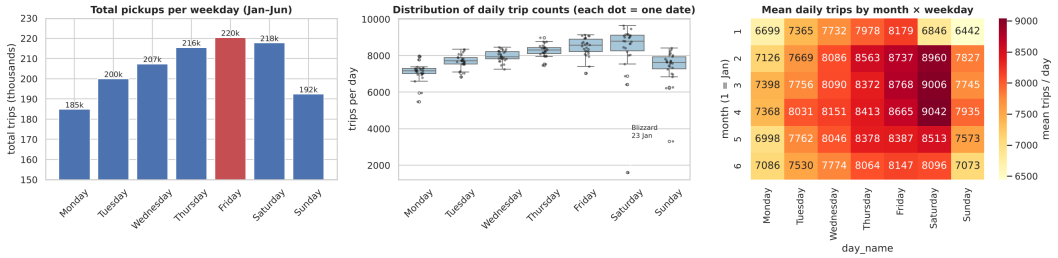


Figure 3: Total pickups per weekday, distribution of daily counts (the blizzard is the outlier), and mean daily trips by month and weekday.

3.2 Most popular hour on each day

Table 2 and Figure 4 show the hourly profile. Weekdays share one shape: an almost empty night (minimum 03:00–04:00), a steep rise at 06:00–08:00 (commute), a plateau, and a maximum in the evening (18:00 Mon/Tue, 19:00 Wed/Fri, 21:00 Thu: commute plus dinner and going out, shifted later as the week's going-out period begins). Saturday and Sunday have no commuter peak: demand grows through the late morning to a broad afternoon plateau and peaks at 23:00 (Saturday) and 00:00

(Sunday), i.e. Friday and Saturday nights spilling past midnight. Weekend nights are much busier (the quietest hour on Fri/Sat/Sun still has 76–87 pickups, against 38–52 on Mon–Thu). Plausible reasons are commuting, dinner/theatre/bar traffic with 4 a.m. closing times driving rides home, and the dip at about 16:00 followed by a jump at 17:00 which is consistent with the taxi shift change (day drivers hand over at 16:00–17:00); between 03:00 and 05:00 only airport runs and shift workers travel.

Table 2: Most popular and quietest pickup hour by weekday (mean trips per hour, storm excluded).

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Peak hour	18	18	19	21	19	23	0
Trips in peak hour	495	512	521	538	538	521	484
Quietest hour	3	3	4	4	4	5	5
Trips in quietest	46	38	47	52	87	76	76

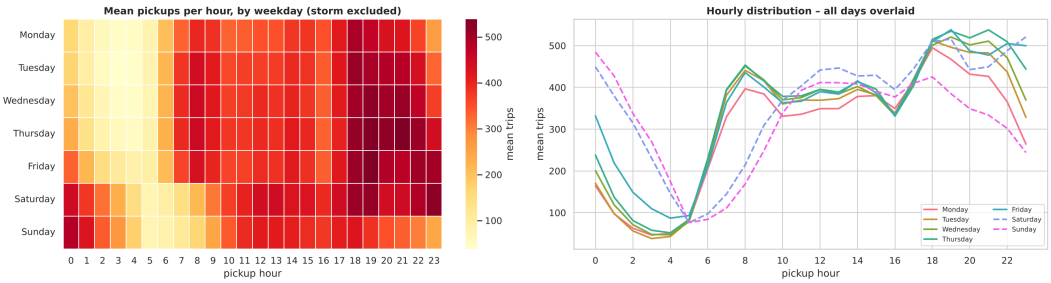


Figure 4: Mean pickups per weekday and hour (heat-map) and hourly profiles of all days.

3.3 Weekdays versus weekends

Total volume is similar (7 923 trips on an average weekday, 8 104 on a weekend day) but the shape differs (Figure 5). 15.3 % of weekday trips start between 07:00 and 10:00 (7.4 % at weekends); the evening 18:00–22:00 share is 25.1 % vs 21.0 %; and 19.8 % of weekend trips start between 00:00 and 05:00 against 7.0 % on weekdays. Weekend trips are faster (15.9 vs 13.9 km/h; the advantage holds between 05:00 and 19:00 but not at night), shorter in time (12.9 vs 14.4 min) for a similar distance (3.5 vs 3.4 km), carry slightly more passengers (1.72 vs 1.64) and start slightly closer to the centre (5.3 vs 5.6 km). The weekday pattern is driven by commuting, congestion and business travel; the weekend pattern by nightlife, brunch, shopping, tourism and events.

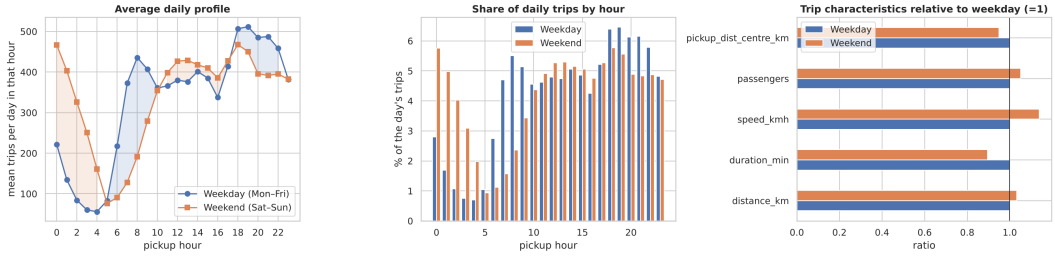


Figure 5: Average daily profile, share of the day's trips by hour, and trip characteristics (weekend relative to weekday).

3.4 Holidays

The 2016 dates are: Martin Luther King Day Mon 18 Jan, Valentine's Day Sun 14 Feb, St Patrick's Day Thu 17 Mar, Easter Sunday 27 Mar and Memorial Day Mon 30 May. Each holiday is compared with all other days of the *same weekday* (excluding other holidays and the storm), using the daily total, the hourly profile, speed and distance (Table 3, Figures 6–7). **Only Memorial Day changes the daily**

Table 3: Holidays versus the same weekday (baseline mean ± sd across 23–25 baseline days).

Holiday	Trips	Baseline	Ratio	z	Peak h (base)	Speed (base)	Dist (base)
MLK Day	7 037	7 231	0.97	-0.6	18 (18)	16.8 (15.0)	3.50 (3.55)
Valentine's Day	8 342	7 542	1.11	$+1.4$	17 (0)	15.8 (16.8)	3.31 (3.77)
St Patrick's Day	8 147	8 294	0.98	-0.4	20 (21)	13.9 (13.5)	3.52 (3.43)
Easter Sunday	7 264	7 542	0.96	-0.5	0 (0)	17.5 (16.8)	3.82 (3.77)
Memorial Day	5 491	7 231	0.76	-5.4	16 (18)	19.1 (15.0)	4.30 (3.55)

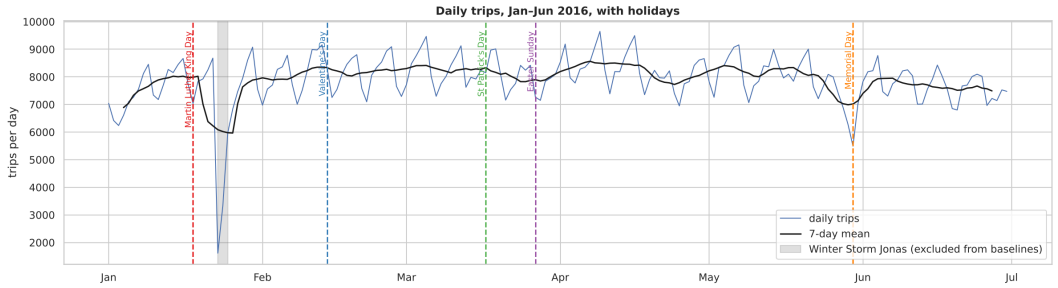


Figure 6: Daily trips with the holidays and the blizzard marked.

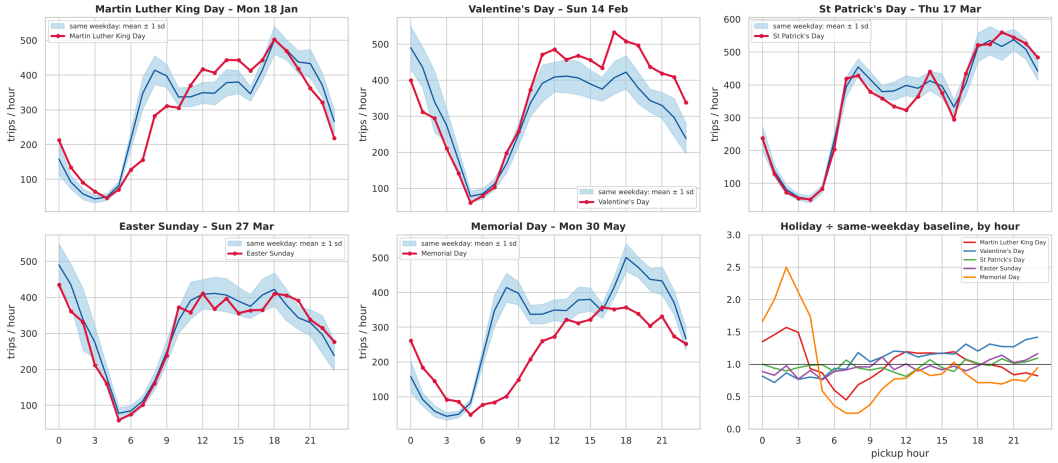


Figure 7: Hourly profile of each holiday (red) versus the same weekday (mean ±1 sd); bottom right: ratio holiday/baseline by hour.

volume significantly: 24 % fewer trips than a normal Monday ($z = -5.4$); the morning commute almost vanishes (07:00 at 24 % of normal), the peak moves from 18:00 to 16:00, after-midnight trips are 2.5× higher at 02:00 (Sunday night of a long weekend), and trips are faster and longer (19.1 vs 15.0 km/h; 4.3 vs 3.55 km) – empty roads and, plausibly, people leaving the city. **MLK Day** has a normal total but a much weaker morning commute (45 % of normal at 07:00), higher speed and busier small hours (1.6× at 02:00). **Valentine's Day** is +10.6 % on a normal Sunday ($z = +1.4$, modest): the peak moves to 17:00 (dinner), the after-midnight spill-over is weaker (0.72× at 01:00) and trips are shorter and slower. **St Patrick's Day** (98 %) and **Easter** (96 %) are close to normal; the largest deviations are a midday dip on 17 March (0.81× at 12:00, plausibly the Fifth Avenue parade closing streets) and slightly higher late-evening demand. Holiday effects are therefore mostly a *redistribution of demand across the hours* rather than a change in volume, with the long-weekend Monday the exception. With one occurrence per holiday these are single-event comparisons.

3.5 Speed through the day

Trips are **fastest at about 05:00** (24.6 km/h on average) and slowest in the middle of the working day (12.1 km/h at 15:00; 12.5 km/h on average between 08:00 and 18:00). Speed rises through the

night from 17.8 km/h at midnight, falls sharply between 06:00 and 08:00 (24.6 to 13.0 km/h) as the rush begins, stays on a flat congested plateau from 08:00 to 18:00 and recovers in the evening (13.6 to 16.9 km/h from 19:00 to 23:00). The 05:00 maximum is partly a trip-mix effect (mean distance 5.5 km against 3.4 km at 15:00: airport and highway trips). Weekends are faster between 05:00 and 19:00. Speeds are haversine-based, so the pattern, not the absolute level, is reliable.

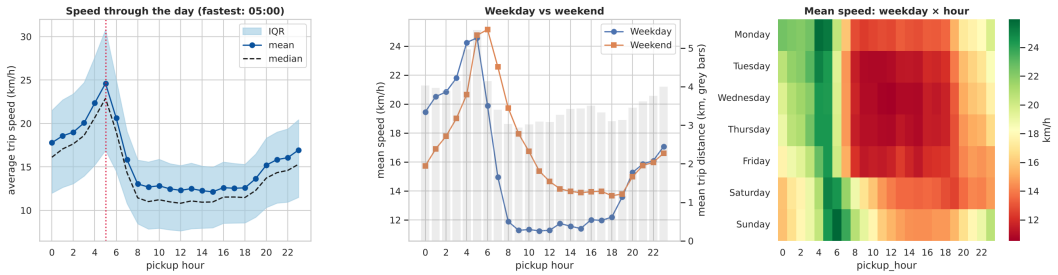


Figure 8: Average trip speed by pickup hour (mean, median, interquartile range), weekday versus weekend, and weekday × hour heat-map.

4 Location clusters

4.1 Heat-maps

Figure 9 shows 2-D histograms (cells of about 100 m×120 m, log colour scale, mean pickups per day) for weekdays/weekends and for morning (07:00–10:00) and evening (18:00–22:00); Figure 10 shows the difference in the share of each group's pickups per cell. The hours were chosen from Section 3: 07:00–10:00 is the commute peak and 18:00–22:00 the weekday maximum. **Findings.** About 93 % of pickups are in Manhattan; the avenue grid is clearly visible, while the other boroughs are sparse and the airports (JFK and LaGuardia, about 2 % each) are isolated spots. Newark has essentially none. *Weekdays vs weekends:* weekday pickups concentrate in Midtown and on the East Side (Midtown within 2 km of Times Square holds 45.0 % of weekday pickups vs 39.9 % at weekends; Grand Central 8.9 % vs 6.4 %; Upper East Side 21.8 % vs 18.1 %), whereas at weekends activity shifts downtown to nightlife districts (East Village/Lower East Side 6.8 % → 10.9 %; the difference map is red over the Lower East Side, East/West Village and Chelsea). *Morning vs evening:* mornings come from residential areas (Upper East Side 26.1 % of morning vs 19.5 % of evening pickups; Penn Station 8.3 % vs 7.1 %), evenings from Midtown (46.4 % vs 42.1 %) and the downtown dining and nightlife areas (7.2 % vs 6.0 %). The spatial pattern therefore mirrors Section 3: a residential-to-business flow in the morning, the reverse in the evening, and a leisure geography at weekends.

4.2 Hotspots

We study Friday 23:00 – Saturday 02:00 and Thursday 17:00 – 20:00, pooled over the 26 weeks (storm excluded): 33 712 and 37 805 pickups in 76 and 78 separate clock-hours, i.e. only 444–485 pickups per hour for the whole city.

The suggested definition cannot be met. A hotspot of "at least 15 pickups within 50 or 100 m in an hour" never occurs: in every single hour of both windows the largest number of pickups within 100 m of one another is 14 (10–11 within 50 m), so the literal rule finds zero hotspots. The training file is a sample of taxi activity and absolute thresholds from the brief are too high. We keep the idea (a persistent *rate* of pickups) and set `min_samples` = (required pickups per hour) × (number of pooled clock-hours), with haversine DBSCAN (ball tree).

Parameters. $\varepsilon = 50$ m: a NYC block is about 80 m×270 m and consumer GPS is accurate to 10–50 m, so 50 m is about one street corner and gives drivers a place they can drive to. With $\varepsilon = 100$ m DBSCAN *chains* neighbouring blocks (the largest cluster has radius 1.2–1.5 km at 1 pickup/hour, i.e. most of Midtown becomes one blob). A rate of **1 pickup per hour** (`min_samples` = 76 and 78) gives compact clusters (largest radius < 200 m); 0.5/hour gives over 100 clusters that begin to chain, 2/hour leaves only 6 (Table 4). **Result: 36 hotspot clusters for Friday 23:00–02:00 and 22 for Thursday 17:00–20:00** (Figure 11; interactive folium maps are in `maps/`). They are compact (radius under 200 m) and contain 18.5 % and 10.4 % of the window's pickups; the rest is noise, so demand is

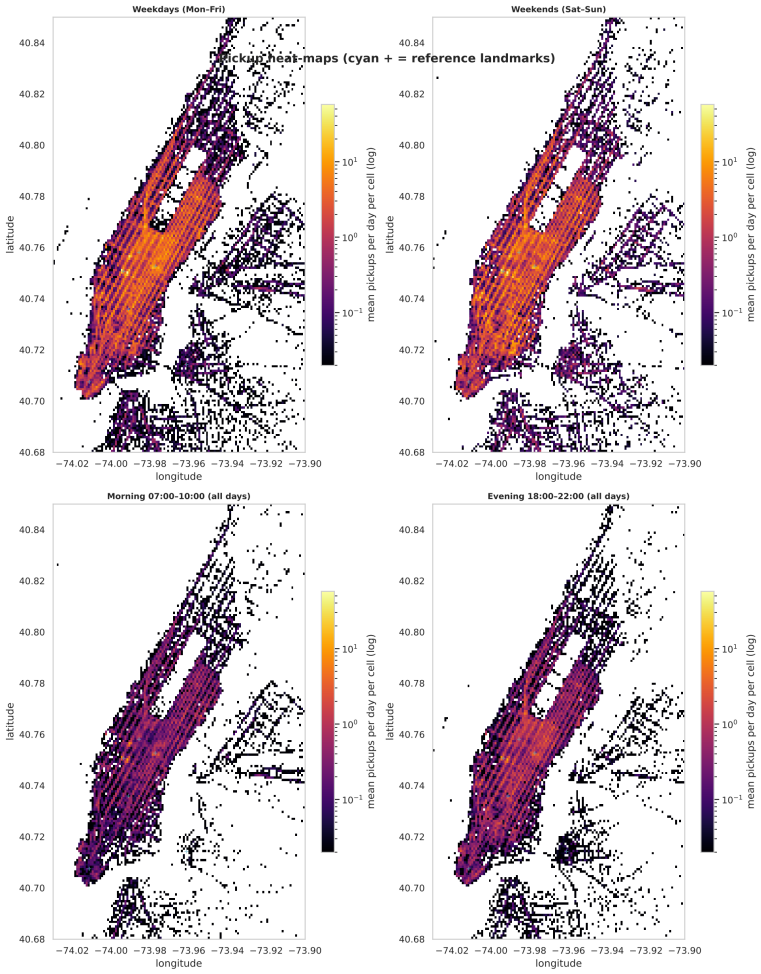


Figure 9: Pickup heat-maps (log density).

Table 4: Number of clusters (largest cluster radius in metres) for different ε and required pickups per hour; Friday night | Thursday evening.

	Fri 23:00–02:00			Thu 17:00–20:00		
ε (m)	0.5/h	1/h	2/h	0.5/h	1/h	2/h
30	77 (183)	12 (60)	0	47 (119)	8 (69)	3 (54)
50	116 (484)	**36 (193)**	6 (77)	115 (603)	**22 (154)**	6 (139)
75	90 (1083)	57 (417)	15 (196)	104 (2586)	43 (743)	7 (188)
100	48 (3289)	47 (1190)	21 (401)	60 (4651)	42 (1502)	18 (292)

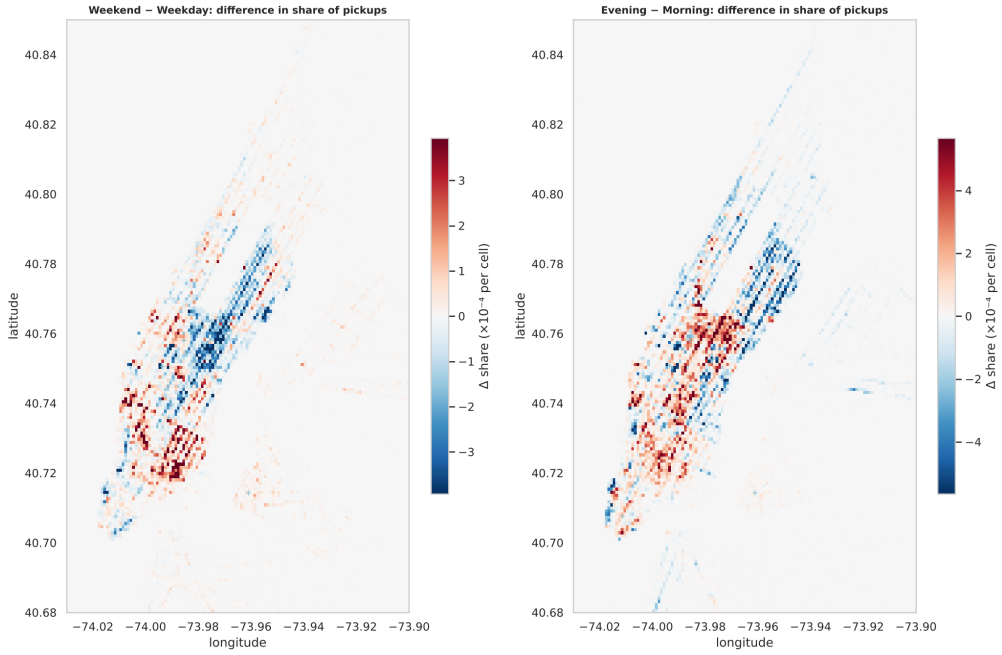


Figure 10: Weekend minus weekday and evening minus morning difference in share of pickups (red = relatively more of the first group).

mostly diffuse. With $\varepsilon = 100\,\text{m}$ and 2/hour the counts would be 21 and 18, so the number is best read as "a few dozen".

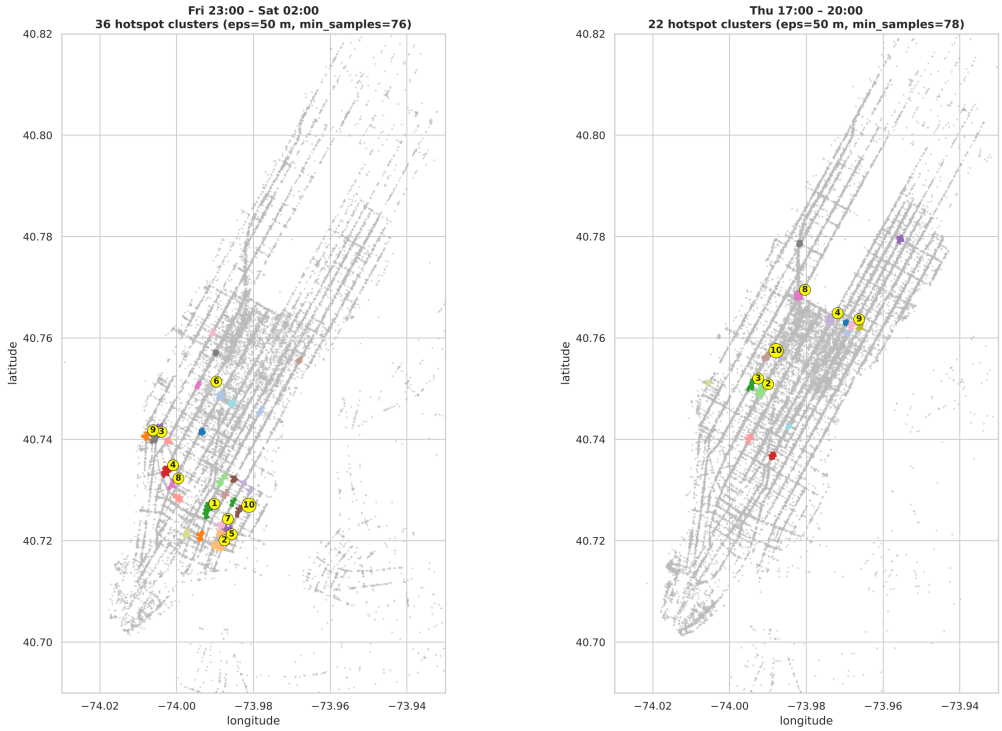


Figure 11: DBSCAN hotspots ($\varepsilon = 50\,\text{m}$, ≥ 1 pickup/hour); grey = noise, numbers = ten busiest clusters (the airport clusters lie outside this map window).

Where are they? (Nearest hand-entered landmark; approximate.) *Friday night* is downtown nightlife: 52.6 % of the clustered pickups lie in the Village/East Village/Lower East Side/Meatpacking area (vs 3.9 % on Thursday), the busiest clusters being around Astor Place/NoHo, Nolita/Bowery on the Lower East Side, the Meatpacking District/West Chelsea (three clusters) and the West Village (2.7–5.2 pickups/hour each), plus Penn Station/Madison Square Garden (3.5/h) and a JFK cluster (2.4/h). *Thursday evening* is transport hubs and the airports: Midtown West (Penn Station, Port Authority, Columbus Circle, Fifth Avenue/57th) holds 40.7 % of clustered pickups (15.9 % on Friday), the Upper East/West Side 21.1 % (0 % on Friday), and the busiest cluster is the LaGuardia terminal area (6.9/h) followed by two Penn Station clusters (5.3 and 3.9/h). Only 7 of the 36 Friday clusters have a Thursday cluster within 250 m, mostly hubs and airports. A driver should therefore wait at the airports and Penn Station/Port Authority/Columbus Circle on Thursday early evening, and in the East Village–Lower East Side–Meatpacking–West Village area after 23:00 on Friday. Limitations: pooling averages out weather and events; a single ε cannot suit both dense Midtown and sparse areas (80–90 % of pickups are noise); airports are hotspots in one direction only.

5 Airports

A trip is an Empire State Building (ESB) $\rightarrow$ airport trip if its pickup lies within R_{ESB} of the ESB and its dropoff within R_{air} of the airport centre (haversine; cleaned data; time of day = pickup hour). The brief labels the points "(long, lat)" but the numbers are (lat, lon) – 40.65 N, 73.78 W is JFK – and are used so. The straight-line distances are ESB–JFK 21.1 km, ESB–Newark 17.6 km.

Radii. Drop-offs around an airport rise steeply with radius to about 1.2 km (JFK) and 1 km (Newark) and then flatten, and the drop-off density per km^2 falls by one to two orders of magnitude at the same radius (Figure 12); $R_{\text{air}} = 2$ km contains the terminal area with a margin for GPS error and service roads (only about 600 more JFK drop-offs than 1.2 km) without including unrelated places. $R_{\text{ESB}} = 0.5$ km (about two long or six short blocks) covers where passengers actually hail cabs (Herald Square, Penn Station, Fifth Avenue) and gives 862 JFK and 236 Newark trips. The result is robust: the median travel time stays between 45.1 and 47.6 min for JFK and between 32.7 and 35.8 min for Newark for every combination of $R_{\text{ESB}} \in [0.2, 1.0]$ km and $R_{\text{air}} \in [1, 3]$ km.

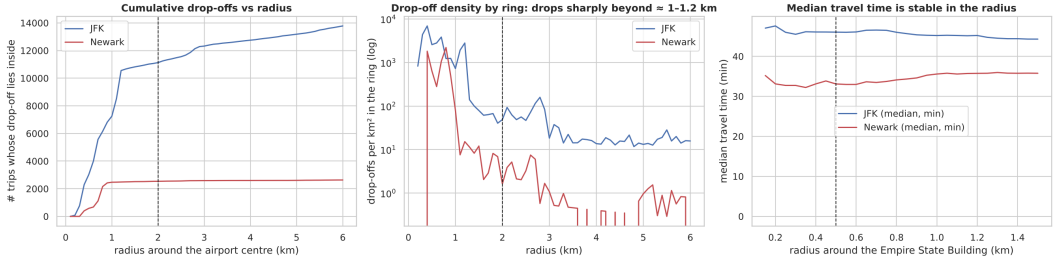


Figure 12: Choice of radii: cumulative drop-offs and ring density around the airports, and median travel time versus the ESB radius.

Table 5: Travel time from the Empire State Building (minutes).

	Trips	Mean	Median	Std	IQR	Distance (km)	Speed (km/h)
JFK	862	49.1	46.0	17.1	36.6–59.9	20.9	28.6
Newark	236	36.3	33.1	11.8	28.3–40.9	17.9	31.8

Results. Travel to JFK takes on average 49.1 min (median 46.0 min); to Newark 36.3 min (median 33.1 min). JFK is about 13 min (35–40 %) slower (Mann–Whitney $p \approx 10^{-31}$): it is about 20 % further and the trips are slower (28.6 vs 31.8 km/h; plausibly the Queens expressways versus a tunnel and New Jersey highways – the routes are not in the data). **By time of day** (Figure 13): to JFK the trip takes about 28–33 min between 01:00 and 05:00, 38–41 min at 07:00–10:00, 43–46 min at 11:00–13:00, then jumps to 58 min at 14:00 and peaks at **66 min at 16:00** (more than twice the night time), falling back to 43 min by 19:00. Period medians for JFK are 28.6 (00–05), 35.4 (06–09), 48.7 (10–15), **59.6** (16–19) and 39.8 min (20–23). Newark has the same shape at a lower level: 26–29

min at 04:00–06:00, 32–37 min in the morning and midday, 42–45 min at 14:00–16:00 and a peak at 17:00 (59 min, but based on only 4 trips); period medians 27.1, 32.8, 34.0, 42.0 and 32.7 min. The afternoon rush costs 2.1× the night time to JFK but 1.55× to Newark. Weekends are faster (JFK median 39.9 vs 50.4 min; Newark 30.6 vs 34.0 min). In practice allow about 45 min to JFK (about an hour in the afternoon rush, 30 min at night) and 33 min to Newark (45 min at 14:00–17:00). Newark has no trips before 04:00 and few after 20:00, and hours with fewer than 5 trips have no confidence interval, so single hours should be read through the confidence bands.

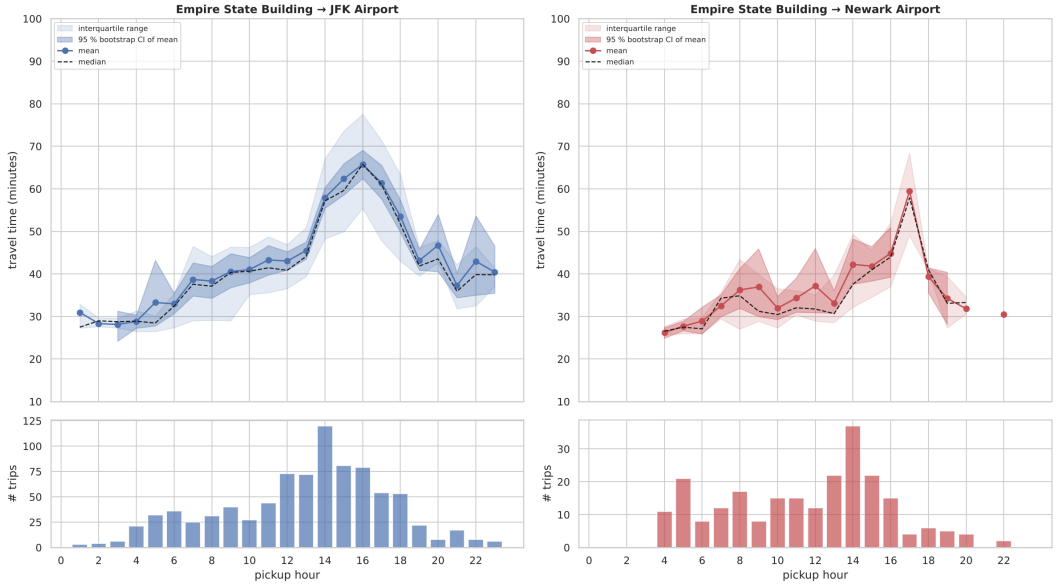


Figure 13: Travel time from the ESB by pickup hour: mean (with 95 % bootstrap CI), median, IQR and number of trips.

6 Dimensionality reduction

Everything uses a random sample of 20 000 cleaned trips (seed 42).

6.1 Feature matrix

The 11 features are: pickup and dropoff longitude/latitude; log distance, log duration and log speed (skewness 2.8, 2.15, 1.46 → 0.37, −0.13, −0.20, so that a few long airport trips do not dominate Euclidean distances); the cyclic hour encoding $(\sin\frac{2\pi h}{24}, \cos\frac{2\pi h}{24})$ so that 23:00 and 00:00 are neighbours; and the same cyclic encoding of the day of week with period 7. All features are standardised (`StandardScaler`), which is essential because PCA, t-SNE, UMAP and DBSCAN are distance/variance based: without it duration in seconds would swamp coordinates in degrees and the unit-circle encodings. A consequence worth stating in advance: day of week takes only seven values, so every trip sits on one of seven fixed points in two of the 11 coordinates, which leaves a strong imprint on the embeddings. The 11th principal component has zero variance because $\log\text{speed} = \log\text{distance} - \log\text{duration} + \text{const}$.

6.2 PCA

The variance is spread out: PC1 explains 22.1 %, PC2 13.9 %, PC3 12.3 %, PC4 10.2 %, so two components capture only 36.0 % and **8 components are needed for 90 %** (83.7 % with 7, 90.2 % with 8; Figure 14). Loadings (Figure 14): **PC1** (22 %) is trip length and speed (distance +0.60, duration +0.46, speed +0.36, pickup longitude +0.35): long, fast trips towards Queens/the airports; **PC2** (14 %) is the uptown–downtown (north-east) position (dropoff latitude +0.61, pickup latitude +0.53); **PC3** (12 %) is time of day (hour_cos +0.61, speed +0.48, duration −0.44: fast short night trips versus slow afternoon trips); **PC4** is almost purely hour_sin (+0.78), the morning–evening

contrast. On the first two components (Figure 15) distance is the only clearly structured colouring: long trips form separate small clouds at PC1 $\approx$ 2.5–3 and 5–6 (the airport groups), whereas hour and weekday/weekend overlap completely because they load on later components. A two-component PCA therefore separates long from short trips and uptown from downtown, but not *when*.

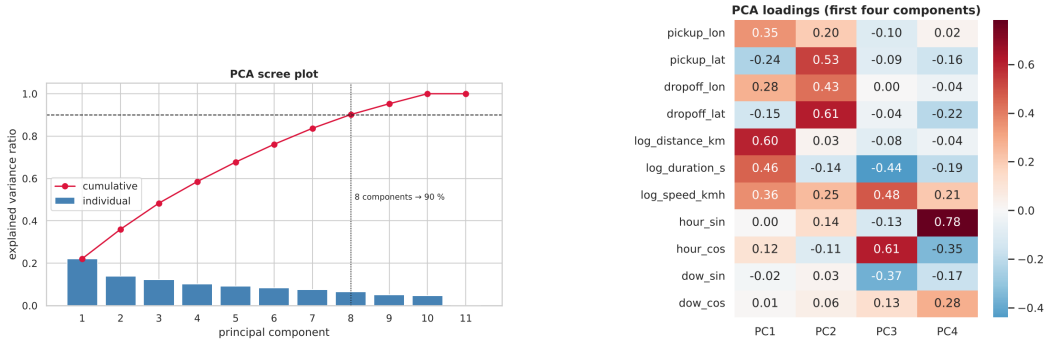


Figure 14: PCA scree plot with cumulative explained variance (left) and loadings of the first four components (right).

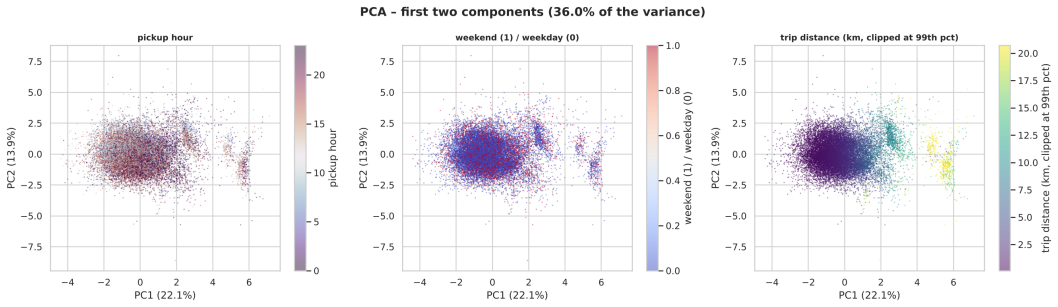


Figure 15: PCA, first two components, coloured by pickup hour, weekday/weekend and trip distance.

6.3 t-SNE

Figure 16 shows t-SNE (PCA initialisation, seed 42) with perplexity 5, 30 and 100. **Perplexity 5** (57 s) is very local: one nearly circular blob without clear groups, with fragmented chains of long trips; it over-emphasises noise. **Perplexity 30** (69 s) shows a flower of about seven large lobes and several small separate islands: colouring by weekend shows that the lobes are the days of the week (two weekend lobes) with the hour changing smoothly inside each, and the islands on the right are the long (airport) trips. **Perplexity 100** (110 s) shows the same groups as fewer, larger, more compact lobes with better separation and cleaner hour gradients, but small structures begin to be absorbed, the cost grows and the relative positions of the lobes change – the distances between islands are not stable, only their membership.

6.4 UMAP

UMAP was run for $n_{\text{neighbors}} \in \{5, 15, 50\}$ and min_dist $\in \{0, 0.5\}$ (Figure 17; the weekend and distance colourings are in the notebook). With $n_{\textbf{neighbors}} = 5$ the groups are fragmented with many small satellite pieces (over-segmentation of noise); with **15** the groups are compact and complete and the airport islands are visible; with **50** the daily groups are most regular, a flower of seven petals, and fine detail is smoothed away: larger $n_{\text{neighbors}}$ trades local detail for a cleaner, more global arrangement. **min_dist** $= 0$ packs points into tight, clearly separated islands with white space between them (ideal for clustering, but it hides density differences inside a group and exaggerates gaps); 0.5 spreads points so that the internal hour gradient is visible but groups touch and boundaries blur. In all six settings the message is the same: seven day-of-week groups (two weekend) with

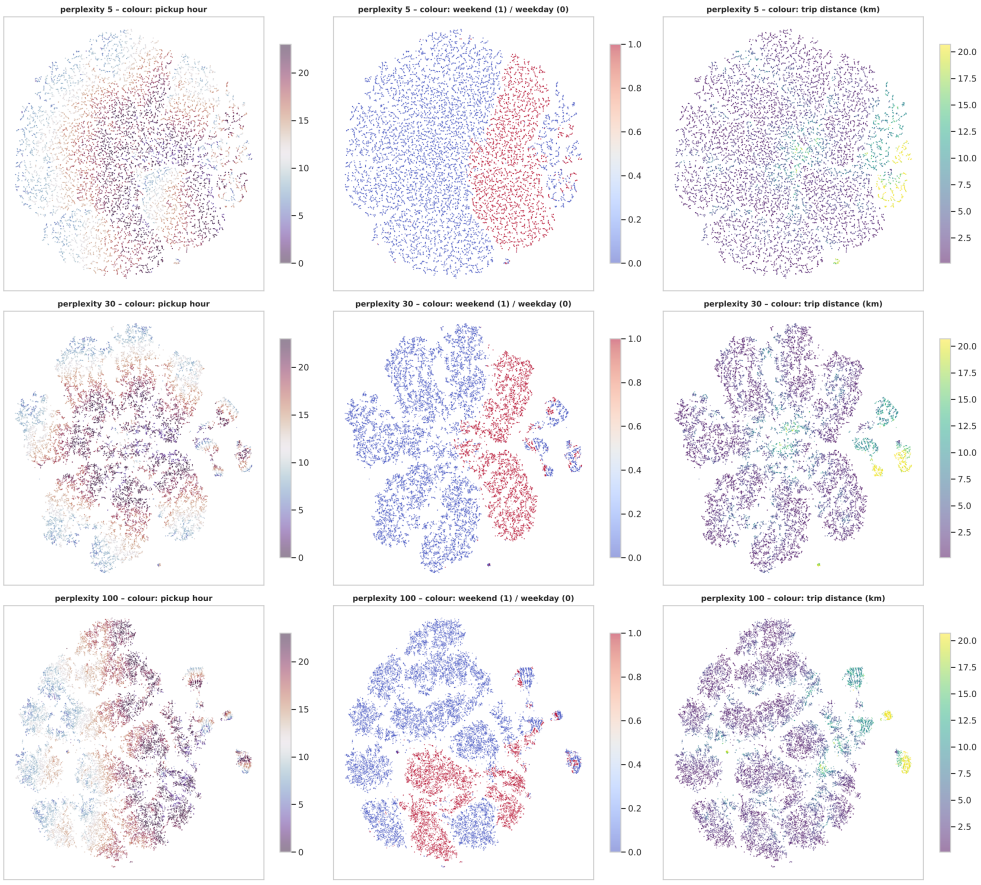


Figure 16: t-SNE with perplexity 5 (top), 30 (middle) and 100 (bottom), coloured by pickup hour, weekday/weekend and trip distance.

smooth hour gradients, and a handful of small groups of long trips. UMAP took 6–30 s per setting (the first includes numba compilation) against 57–110 s for t-SNE.

6.5 Density-based clustering and cluster profiles

Best embedding. We choose the best t-SNE/UMAP embedding quantitatively with the *trustworthiness* (fraction of each point's 15 nearest embedded neighbours that are also near neighbours in the original space; 4 000-trip subset) and the Spearman correlation of pairwise distances (global structure) (Table 6): t-SNE perplexity 100 (0.980, just ahead of perplexity 30 with 0.977) and UMAP $n_{\text{neighbors}} = 50$, min_dist $= 0$ (0.960) are selected. **Method and parameters.** DBSCAN with

Table 6: Quality of the 2-D embeddings.

Embedding	Trustworthiness ($k = 15$)	Distance correlation
PCA, 2 components	0.768	0.750
t-SNE perplexity 5 / 30 / 100	0.945 / 0.977 / **0.980**	0.323 / 0.329 / 0.405
UMAP $n = 5$, md 0 / 0.5	0.956 / 0.918	0.223 / 0.220
UMAP $n = 15$, md 0 / 0.5	0.954 / 0.929	0.286 / 0.304
UMAP $n = 50$, md 0 / 0.5	**0.960** / 0.929	0.363 / 0.386

`min_samples`$= 30$ for all methods (a cluster must contain at least 30 trips, 0.15 % of the sample) and ε from the knee of the sorted 30-th-neighbour distance plot (Figure 18): $\varepsilon = 1.42$ (PCA), 2.53 (t-SNE), 0.16 (UMAP); they differ because the spaces have different scales. PCA is clustered in the 8 dimensions that explain 90 % of the variance and visualised on the first two components; t-SNE and UMAP are clustered on the 2-D embeddings. **PCA** finds one giant cluster (88.7 %: median 1.95

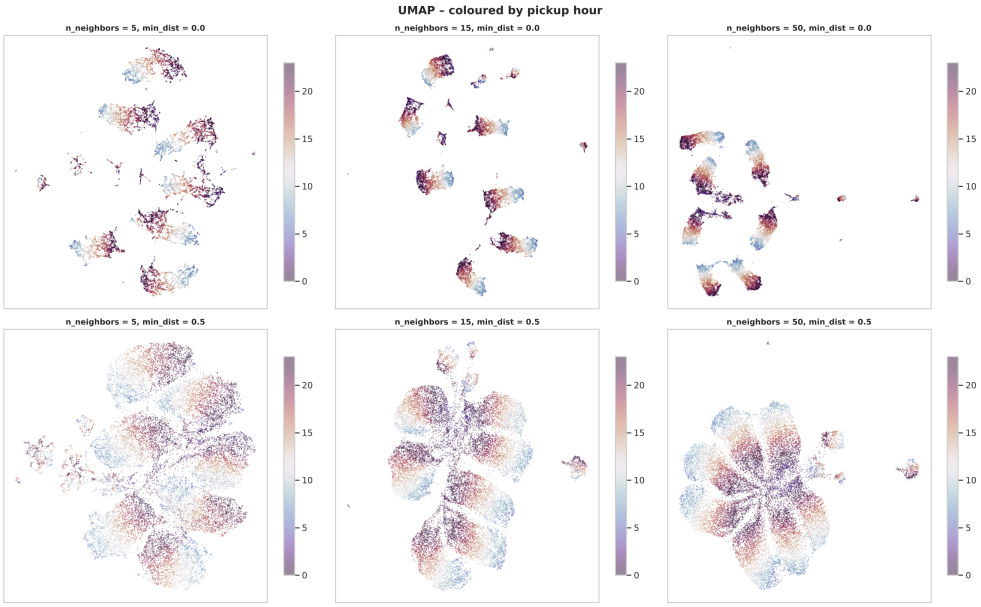


Figure 17: UMAP embeddings coloured by pickup hour: columns $n_{\text{neighbors}} = 5, 15, 50$, rows min_dist $= 0, 0.5$.

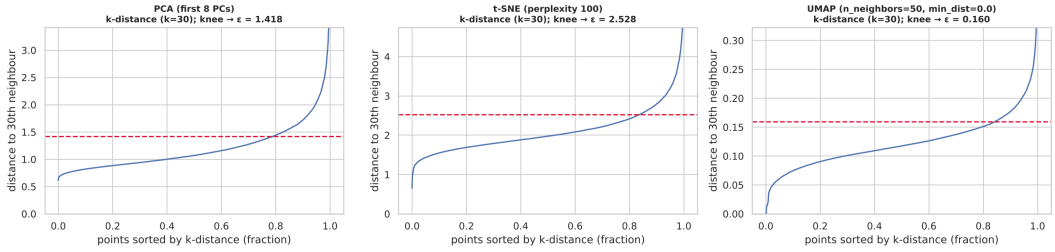


Figure 18: k-distance plots (k=30) with the chosen ε (dashed).

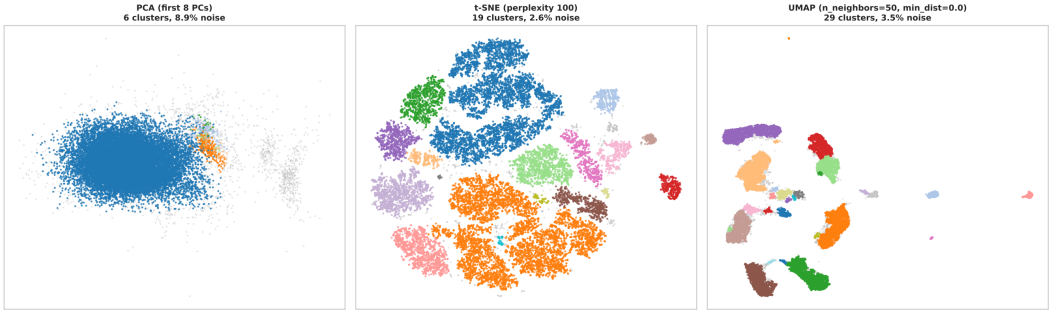


Figure 19: DBSCAN clusters (noise grey) in the three spaces.

Table 7: Clustering results.

Method (space)	ε	Clusters	Noise	Largest cluster
PCA (8 PCs)	1.42	6	8.9 %	88.7 %
t-SNE (perplexity 100)	2.53	19	2.6 %	29.1 %
UMAP (50 neighbours, min_dist 0)	0.16	29	3.5 %	12.9 %

km, 10.3 min, 12.2 km/h, spread over the day and week) plus a few small airport clusters (23–246 trips); its noise group is half airport trips (median distance 10.5 km). So PCA with density clustering separates only "airport/long" from "everything else". **t-SNE** (19 clusters) isolates Thursday midday and evening, Friday midday, Tuesday and Wednesday mornings and Sunday late night, but merges Saturday and Sunday (and some weekday trips) into one cluster (29.1 %) and several weekdays into another (26.4 %); six small clusters are airport runs. **UMAP** (29 clusters) finds each day of the week as its own cluster (Monday, Tuesday, Wednesday, Friday, Saturday, Sunday each 8–13 % of the sample and 99–100 % one day; Thursday split into a midday and an evening cluster; Sunday also has a small late-night cluster, i.e. Saturday night after midnight) plus small clusters of long trips. Table 8 profiles the main UMAP clusters on the original (unscaled) features; the names come from a fixed rule applied to these statistics (time of day from the circular mean hour; weekday/dominant day; length from the median distance; "airport runs" if more than 50 % of the trips start or end within 2 km of JFK, LaGuardia or Newark), so they are reproducible and consistent with the numbers. **Interpretation.** The great majority are short (1.7–2.0 km, 9–11 min, 10–14 km/h) Midtown/Grand

Table 8: Profile of selected UMAP clusters (original features; hour = circular mean; "airport" = share of trips within 2 km of JFK/LGA/Newark).

Name	Cl.	%	Dist (km)	Dur (min)	Speed	Hour	Weekend	Airport
all-day Fri short trips	8	12.9	1.79	10.0	11.1	16.6	0.00	0.00
all-day Sat short trips	3	12.8	1.90	9.4	12.7	17.5	1.00	0.00
all-day Wed short trips	2	12.1	1.77	10.4	10.7	16.2	0.00	0.00
all-day Tue short trips	4	11.8	1.75	9.7	11.3	15.8	0.00	0.00
all-day Mon short trips	10	10.2	1.73	9.1	12.1	15.8	0.00	0.00
midday Sun short trips	11	8.4	1.94	8.7	13.8	14.8	1.00	0.00
midday Thu short trips	6	6.3	1.78	10.9	10.1	10.3	0.00	0.00
evening Thu short trips	5	6.1	1.97	10.1	12.1	20.2	0.00	0.00
late-night Sun short trips	13	1.9	1.81	8.1	13.7	1.4	0.99	0.00
late-night mixed-week long trips	0	1.3	9.74	19.9	29.3	23.4	0.37	0.01
midday mixed-week long airport runs (LGA)	1	2.3	9.76	28.5	20.6	16.0	0.20	0.98
all-day mixed-week very long airport runs (JFK)	7	2.1	19.83	40.2	27.2	17.8	0.25	0.97
midday mixed-week very long airport runs	12	0.8	20.47	42.8	28.7	13.2	0.37	0.88

Central/Times Square hops spread over the day, one cluster per day of the week; their speeds reflect the congestion of Section 3.5 (Wednesday slowest at 10.7 km/h, Saturday fastest at 12.7). Thursday splits into the slowest cluster (midday, 10.1 km/h) and an evening cluster. Late-night Sunday (i.e. Saturday night) trips are short and quick (1.4 h mean hour); a fast, long late-night group (9.7 km at 29.3 km/h) corresponds to highway or cross-town night rides. The airport runs form distinct clusters: LaGuardia-area trips (about 9.8 km, 28.5 min) and JFK trips (about 20 km, 40–43 min, 27–29 km/h; two clusters that are probably the two directions of the same route, one centred at JFK and one in Midtown). The maps in the notebook (and appendix Figure 20) show that the day-of-week clusters cover the same Manhattan footprint, so location does not separate them; only the airport clusters are spatially distinct.

6.6 Comparison of PCA, t-SNE and UMAP

Consistent groups. *Airport trips form their own group in all three methods*: in t-SNE and UMAP they are compact islands separated from the bulk (UMAP: three well separated islands plus smaller ones), in PCA they lie away from the bulk along the long-trip direction but as a diffuse band overlapping its edge. The seven day-of-week groups and the late-night weekend groups appear in t-SNE and UMAP but not in PCA. **PCA** reveals the dominant linear axes (interpretable through the loadings), preserves global distances best (distance correlation 0.75 vs 0.41 for t-SNE and 0.36 for UMAP) and takes well under a second, but it misses local structure (trustworthiness 0.77 against 0.98 and 0.96) and cannot resolve the day/time groups or the small airport groups.

Local versus global structure. t-SNE and UMAP preserve neighbourhoods far better than PCA (trustworthiness 0.98/0.96 vs 0.77; UMAP is better with min_dist $= 0$) but preserve global distances poorly (0.2–0.4 vs 0.75). *Distances between clusters should not be interpreted*: the arrangement of the lobes changes with the perplexity and with the seed (Figure 22), and the gaps are largely created by the algorithm (the silhouette of the clusters in the original space is about zero, below). *Cluster sizes are not interpretable either*: t-SNE equalises point density and UMAP with min_dist $= 0$ packs points tightly, so the area of an island does not measure its population or spread. Only membership, and to a lesser degree local neighbourhoods, can be read.



Figure 20: Pickup locations (red) of the eight largest and the airport-dominated UMAP clusters; grey = all sampled pickups.

Table 9: Runtime and sensitivity (20 000 trips, 4 cores).

	Runtime	Hyper-parameters	Seed
PCA	well under 1 s	only number of components	deterministic
t-SNE	57 / 69 / 110 s (perplexity 5/30/100); 93–116 s with random init	high: perplexity 5 gives no groups; 30 and 100 similar but lobes arranged differently	random init: 21–24 clusters, ARI 0.85 / 0.45 / 0.39 between seeds; with PCA init nearly seed-independent (ARI 1.0, trivially)
UMAP	6–44 s (first call includes numba compilation)	high: fragmentation and gap size change with $n_{\text{neighbors}}$, min_dist	29 / 20 / 23 clusters; ARI 0.86, 0.86, 0.90

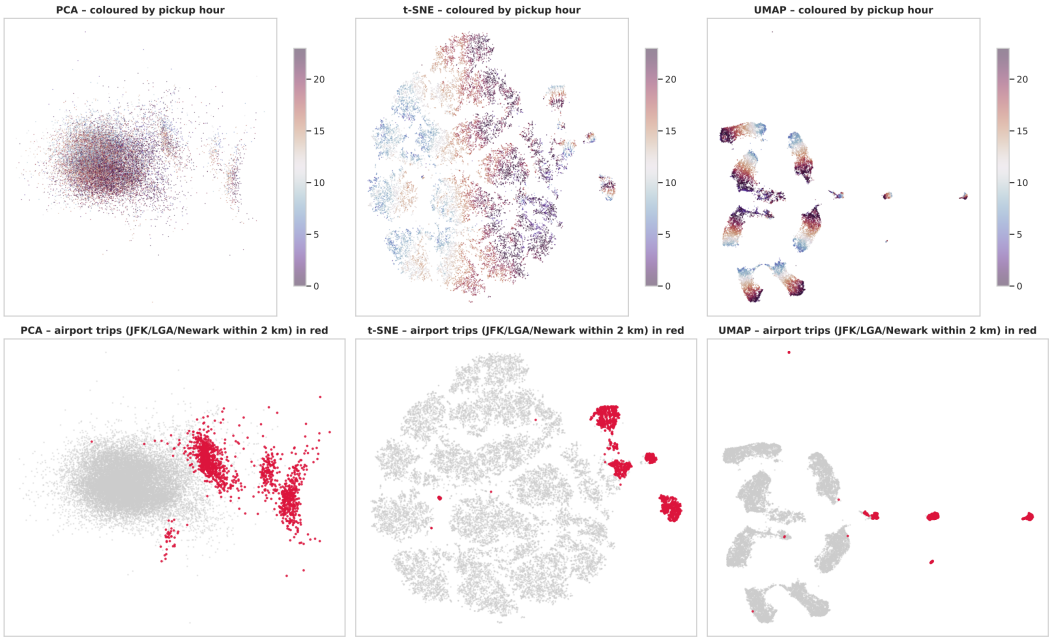


Figure 21: Best embedding of each method side by side; top: coloured by pickup hour; bottom: airport trips (pickup or dropoff within 2 km of JFK, LaGuardia or Newark; 6.7 % of the sample) in red.

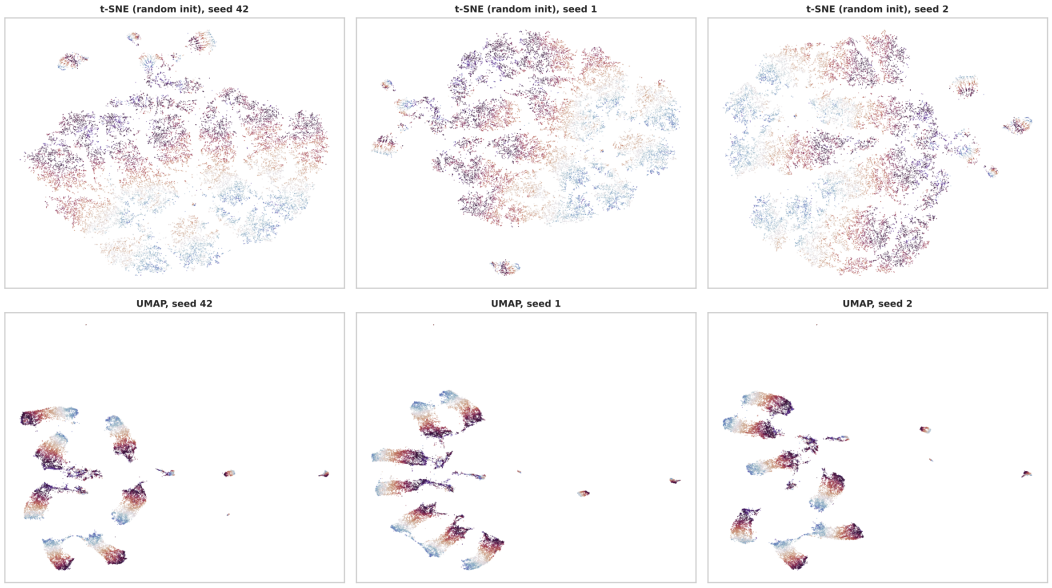


Figure 22: Seed sensitivity: best t-SNE (random initialisation) and best UMAP for seeds 42, 1 and 2 (colour: hour).

Agreement and trust. The adjusted Rand index (ARI) between the clusterings is 0.42 for t-SNE vs UMAP (0.45 on the trips clustered by both), but only 0.08 for PCA vs t-SNE and 0.04 for PCA vs UMAP (0.03 and 0.01 on the trips clustered by both): t-SNE and UMAP agree on the airport, single-day and late-night groups but differ in where they merge neighbouring days, and the PCA clustering (one cluster of 88.7 % plus airport groups) cannot agree with the 19–29 day/time clusters; ARI also penalises the different number of clusters. Can embedding clusters be trusted as real groups? *Partly.* A kNN classifier working only on the original 11 standardised features recovers the t-SNE and UMAP cluster labels with 95.9 % and 96.2 % cross-validated accuracy (majority baselines 30.3 % and 13.4 %), so the clusters are consistent with neighbourhoods in the original space and are not artefacts of the optimiser; the airport clusters are confirmed independently by Section 5. But the silhouette of these clusters computed in the original space is -0.011 (t-SNE) and 0.000 (UMAP): there is no empty space between day-of-week groups there (they differ in two of 11 coordinates), the visible gaps are produced by the embedding, the number of clusters changes with seed and parameters (t-SNE 21–24, UMAP 20–29) and the airport trips are split into several clusters (the largest holds only 33 % (t-SNE) and 34 % (UMAP) of them). Clusters from t-SNE or UMAP are hypotheses to be validated by profiling in the original space, by repeating with other seeds and parameters, and by independent knowledge.

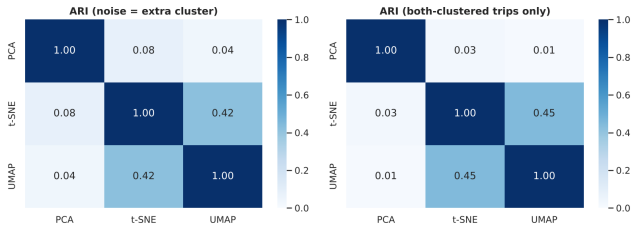


Figure 23: Adjusted Rand index between the three clusterings.

Recommendation. For exploring this data set we recommend **UMAP as the primary tool** (small min_dist, $n_{\text{neighbors}} \approx 15$–$50$), **PCA alongside** for interpretable axes, and t-SNE only as a cross-check. UMAP was several times faster than t-SNE, more stable across seeds in our test (ARI 0.86–0.90 against 0.39–0.85 for random-initialisation t-SNE), nearly as locally faithful (0.96 vs 0.98) with comparable global fidelity, and its separated islands make density-based clustering straightforward. PCA is deterministic and instantaneous, shows that eight dimensions are needed for 90 % of the variance and has readable loadings; every cluster found in UMAP or t-SNE should still be profiled in the original feature space. A caveat on our own results: the seven day-of-week islands arise largely because day of week is a discrete feature in the feature matrix; encoding it as a weekend flag (or omitting it) would remove them and emphasise location, distance and speed.

Conclusions

The cleaned data (98.6 % of trips) show a clear weekly rhythm: Friday/Saturday are busiest, weekdays are commute-driven (peaks 18:00–21:00) and weekends nightlife-driven; only Memorial Day alters the daily volume. The spatial pattern is a residential-to-business flow in the morning, the reverse in the evening, and downtown nightlife at weekends; taxi hotspots are compact street corners that differ between Friday night and Thursday evening. Trips from the Empire State Building take about 46 min to JFK and 33 min to Newark, with a strong afternoon penalty. Dimensionality reduction shows that the data are high-dimensional (8 PCs for 90 %); the non-linear methods expose the day-of-week and airport structure, but their clusters, distances and sizes must be validated rather than read at face value.